

Session - 1

Java Lambda Expressions:

```
import java.io.*;

import java.util.*;

interface PerformOperation {

    boolean check(int a);

}

class MyMath {

    public static boolean checker(PerformOperation p, int num) {

        return p.check(num);

    }

    public PerformOperation isOdd() {

        return (int a) -> a % 2 != 0;

    }

    public PerformOperation isPrime() {

        return (int a) -> {

            if (a < 2) return false;

            for (int i = 2; i <= Math.sqrt(a); i++) {

                if (a % i == 0) return false;

            }

            return true;

        };

    }

    public PerformOperation isPalindrome() {

        return (int a) -> {

            String original = Integer.toString(a);

            String reversed = new StringBuilder(original).reverse().toString();

            return original.equals(reversed);

        };

    }

}
```

```
public class Solution {  
    public static void main(String[] args) throws IOException {  
        MyMath ob = new MyMath();  
        BufferedReader br = new BufferedReader(new InputStreamReader(System.in));  
        int T = Integer.parseInt(br.readLine());  
        PerformOperation op;  
        boolean ret = false;  
        String ans = null;  
        while (T--> 0) {  
            String s = br.readLine().trim();  
            StringTokenizer st = new StringTokenizer(s);  
            int ch = Integer.parseInt(st.nextToken());  
            int num = Integer.parseInt(st.nextToken());  
            if (ch == 1) {  
                op = ob.isOdd();  
                ret = ob.checker(op, num);  
                ans = (ret) ? "ODD" : "EVEN";  
            } else if (ch == 2) {  
                op = ob.isPrime();  
                ret = ob.checker(op, num);  
                ans = (ret) ? "PRIME" : "COMPOSITE";  
            } else if (ch == 3) {  
                op = ob.isPalindrome();  
                ret = ob.checker(op, num);  
                ans = (ret) ? "PALINDROME" : "NOT PALINDROME";  
            }  
            System.out.println(ans);  
        }  
    }  
}
```

Output:

Congratulations

You solved this challenge. Would you like to challenge your friends?  

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✓ Test case 0

2 1 4

✓ Test case 1 

3 2 5

4 3 898

✓ Test case 2 

5 1 3

6 2 12

Expected Output

1 EVEN

2 PRIME

3 PALINDROME

4 ODD

5 COMPOSITE

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Java Comparator:

```
import java.util.*;
```

```
class Checker implements Comparator<Player> {
```

```
    @Override
```

```
    public int compare(Player a, Player b) {
```

```
        if (a.score == b.score) {
```

```
            return a.name.compareTo(b.name);
```

```
        } else {
```

```
            return Integer.compare(b.score, a.score);
```

```
        }
```

```
    }
```

```
}
```

```
class Player{
    String name;
    int score;
    Player(String name, int score){
        this.name = name;
        this.score = score;
    }
}

class Solution {
    public static void main(String[] args) {
        Scanner scan = new Scanner(System.in);
        int n = scan.nextInt();
        Player[] player = new Player[n];
        Checker checker = new Checker();
        for(int i = 0; i < n; i++){
            player[i] = new Player(scan.next(), scan.nextInt());
        }
        scan.close();
        Arrays.sort(player, checker);
        for(int i = 0; i < player.length; i++){
            System.out.printf("%s %s\n", player[i].name, player[i].score);
        }
    }
}
```

Output:

Congratulations

You solved this challenge. Would you like to challenge your friends?  

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✓ **Test case 0**

Compiler Message

✓ **Test case 1** 

Success

✓ **Test case 2** 

Input (stdin)

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✓ **Test case 3** 

```
1 5
2 amy 100
3 david 100
4 heraldo 50
5 aakansha 75
6 aleksa 150
```

✓ **Test case 4** 

✓ **Test case 5** 

Expected Output

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